TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Automated 3D hiPSC Liver Model Platform for High-Precision Toxicity Testing*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		*Bio Medical*	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	*Bio-process Technologies*	*Industrial Biotechnology*	*Other Industrial Biotechnology*
Concept of Proposed Technology	The German–Korean R&D initiative aims to combine hiPSC-derived 3D liver biology with advanced automation, robotics, and artificial intelligence. The partnership is set to produce a TRL-5/6 prototype that offers reliable, high-throughput DILI assessment for the pharmaceutical and biotechnology sectors. By developing automation modules, the project also supports the global competitiveness of Korean SMEs. Additional benefits include creating mutual value through cross-validation, harmonised standards, and collaborative innovation pipelines. This approach speeds up industrialisation and minimises manual work and variability by implementing comprehensive automation solutions. The project aims to develop a **standardized, scalable, physiologically relevant 3D liver toxicity testing platform** based on **hiPSC-derived multicellular liver aggregates**. A key feature is			

	the **production of aggregates inside a controlled suspension bioreactor**, enabling: • homogeneous aggregate formation, • reproducible culture conditions, • scalable microtissue production. The **Korean partner** will develop a **fully automated assay workflow**, including: • automated aggregate withdrawal from the bioreactor, • controlled handling of aggregate suspensions, • precise deposition into 96-well plates, • robotic toxicity assay execution and **AI-based image analysis**. This combined platform enables reproducible, human-relevant **drug-induced liver injury (DILI)** assessment with true high-throughput compatibility. **This proposal outlines the core concept and a potential project roadmap, which will be further developed alongside the participating Korean SME.**
Rationale for Joint R&D Project	**Fraunhofer IBMT** contributes: hiPSC expansion, differentiation, aggregate generation, biological validation logistics of cell models, biosensor technology. **Korean partner** contributes: automation engineering, microfluidics, robotic handling, imaging & AI. **Impact for Korea & Global Competitiveness** • **Direct Impact for Korea's strategic tech fields:** The automation, robotics and AI workstreams strengthen Korea's national priorities (Advanced Bio, AI, Advanced Manufacturing) with prototypes and know-how spillover to SME suppliers (robotics modules, microfluidic cartridges, imaging analytics). This supports the Korean SMEs' scale-up readiness and product portfolio. • **Industrialization pathway for Korean SMEs:** By building the assay automation chain in Korea and validating it with biological models developed by the German applied research institute, Korean SMEs can position themselves as solution providers to pharma CROs and biotech firms globally, accelerating **export potential** and **global market penetration**. • **Joint IP & standard setting:** Binational results (interfaces, QC metrics, AI models) can be advanced toward de-facto standards, improving Korea's **global competitiveness** in bioprocess automation platforms. **Synergies Between German and Korean Funding Systems** • **Germany (Fraunhofer IBMT):** leverages public research infrastructure, GLP-oriented labs and translational frameworks for human-relevant models, enabling **pre-competitive biological assets**. • **Korea (TIPA & SME programs):** enable **rapid prototyping and productization** on the automation/AI side with short iteration cycles typical for SME-driven innovation. • **Synergy Mechanism:** German public research funding de-risks biology (reproducible

aggregates, validated endpoints), while Korean SME funding accelerates **product engineering** (hardware modules, control software). The combined pipeline shortens time-to-prototype and creates **commercially exploitable modules** on the Korean side and **validated application use-cases** on the German side.

Why Neither Side Can Achieve the Goals Alone

- **Biology ≠ Automation:** High-fidelity hiPSC liver aggregates require deep stem-cell and tissue-engineering expertise not typically housed in automation SMEs. Conversely, building reliable bioreactor interfaces, precision liquid handling and AI analytics at scale requires specialized mechatronics and software engineering not resident in biology institutes.
- **Cross-environment validation:** Only a cross-country setup can prove robustness across different lab settings and supply chains—critical for later adoption by global pharma users. A single-country project would risk **context-specific solutions** that fail to generalize.
- **Market access & customer proximity:** Germany provides proximity to EU pharma/biotech R&D hubs; Korea provides an agile automation supplier base. Without the other, each side lacks either **credible biological validation** (for automation) or **scalable industrial execution** (for biology).

Risks if the Collaboration Does Not Take Place

- **Delayed industrialization:** Without binational co-development, automation remains a lab prototype or the biology remains non-automated—both delaying **TRL 5–6** integration and market entry.
- **Lower reproducibility & acceptance:** Absence of cross-site validation weakens evidence on robustness, slowing **regulatory and industry acceptance**.
- **Missed standardization window:** Competing international consortia may set interfaces and QC benchmarks first; late entry reduces **global competitiveness** for Korean SMEs and diminishes return on public funding.
- **Inefficient use of funds:** Parallel, uncoordinated national projects duplicate effort and increase risk of non-compatible solutions.

Key Words	hiPSC 3D Liver Model, DILI Screening, Automated Toxicity Assay, Bioreactor-derived Aggregates, High-Throughput Robotics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3) laboratory proof-of-concept	End of the Project	Level (5-6) integrated automated prototype
R&D Objectives	• Develop a **functional hiPSC-derived 3D liver model** for toxicity screening. • Produce aggregates in a **controlled, senzorized suspension bioreactor**. • Create an **automation pipeline** for aggregate withdrawal, handling, and dispensing. • Integrate automated imaging, acquisition and **AI-supported toxicity classification**.			
R&D Target Specifications	• Robust 3D liver aggregates with key liver functions. • Automated recovery, compatible with robotics. • Dispensing of 1–3 aggregates/well with <10% variation. • High throughput capability: ≥96 assay conditions/run. • AI based classification of toxicity.			

R&D Plan	**Phase I – Pre-Study (6 Months)**
	Title: Pre-Validation Study for Automated 3D hiPSC Liver Aggregates and Assay Workflow
	Objective: Reduce risks and ensure readiness for Phase II through limited pilot studies.
	
	• **WP 1 – Biological Feasibility (IBMT)**
	  ○ Selection of hiPSC lines
	  ○ Validate hiPSC expansion and basic differentiation in suspension bioreactors
	  ○ Confirm initial aggregate formation parameters
	  ○ Assess viability and basic functional markers
	• **WP 2 – Automation Feasibility (Korean Partner)**
	  ○ Test minimal prototype for aggregate transfer & suspension handling
	  ○ Verify pipetting accuracy and 96-well compatibility
	• **WP 3 – Joint Definition of Phase II Specifications**
	  ○ Align biological and automation needs
	  ○ Define bioreactor–robotics interfaces
	  ○ Joint risk assessment and Phase II planning
	
	Expected Outcomes of Phase I
	  ○ Feasibility confirmed for aggregates and baseline automation
	  ○ Key process parameters identified
	  ○ Shared roadmap for Phase II
	
	Phase II (3-Year Main Project)
	• **WP 1 – Advanced Biology Development (Lead: IBMT, Year 1–2)**			

	○ Optimize differentiation toward mature liver phenotypes ○ Evaluate integration of biomaterials or other cell types ○ Integrate necessary biosensors in bioreactor to monitor the production process ○ Validate functional responses to hepato-toxins • **WP 2 – Automation Architecture & Development (Lead: Korean SME, Year 1–2)** ○ Engineer bioreactor interface modules ○ Develop suspension-analysis and precision dispensing units ○ Build robotic 96-well plating workflow • **WP 3 – Imaging, Data Processing & AI Analytics (Lead: Joint, Year 2)** ○ Implement high-content imaging pipeline ○ Develop AI-based toxicity classification models ○ Integrate QC tracking and feedback loops • **WP 4 – System Integration & Cross-Site Validation (Lead: Korean SME, Year 2–3)** ○ Integrate biological model and automation modules ○ Conduct cross-country validation (Germany ↔ Korea) ○ Perform pilot toxicity screening campaigns • **WP 5 – Logistics & Sample Exchange (Lead: IBMT, Year 1–3)** ○ Standardize international sample shipment procedures ○ Implement a fundamental QC for shipments ○ Define controlled-temperature transport workflows • **WP 6 – Prototype Finalization & Industrialization Roadmap (Lead: Joint, Year 3)** ○ Finalize TRL-5/6 integrated prototype ○ Prepare scale-up documentation ○ Define Korean SME market-entry strategy

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• hiPSC culture & differentiation: generation of stable hiPSC seed banks and production of differentiation batches. • Aggregate production in bioreactors: routine operation of suspension bioreactors; optimization of seeding densities and feeding strategies, etc. • Biological QC assays: e.g., viability, LDH/ATP release, albumin secretion, CYP activity panels. • Toxicity reference testing: validation runs with selected compounds to benchmark assay sensitivity. • Sample logistics coordination: preparation, packaging, and shipment of biological material to Korea (frozen aggregates or single cells). • Integration support: providing biological acceptance criteria for automated modules.
Expected R&R of Korean SME Partner	• Automated harvesting system design: development of a robotic arm tool for gentle aggregate withdrawal. • Robotic suspension handling: creation of microfluidic channels or pipetting units that maintain aggregate integrity. • Automated 96-well plating: precision placement ensuring 1–3 aggregates/well • High-content imaging & data acquisition: setting up automated microscopes with standardized exposure/focus routines. • AI-based toxicity evaluation: training machine-learning classifiers to detect morphological and functional toxicity signatures. • System integration: combining mechanical, electronic, and software components into a cohesive workflow compatible with IBMT's biological models.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **Scientifically**: delivers a reproducible, human-relevant 3D liver toxicity system improving DILI prediction. • **Technologically**: establishes an integrated TRL-5/6 prototype combining bioreactor biology, automation, robotics, and AI. • **Economically**: enables export-ready automation modules and strengthens Korean SME competitiveness in global pharma markets. • **Bilaterally**: creates a sustainable German–Korean innovation pipeline supported by shared standards and cross-validation. • **Industrially**: reduces assay variability and manual labor through end-to-end automated workflows, supporting high-throughput drug testing.

6. References

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